VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-07 PHYSICS

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Topics To Be Covered

1. ELECTROMAGNET
2. ELECTRIC BELL
3. FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD
4. QUESTIONS


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USE OF ELECTROMAGNET IN AN ELECTRIC BELL

An **electric bell** is one of the most commonly used application of an **electromagnet**.

Construction and Wiring:

The main parts of the bell are:

(i) a horse-shoe electromagnet M, having a soft iron core,

(ii) a soft iron armature A,

(iii) a hammer H,

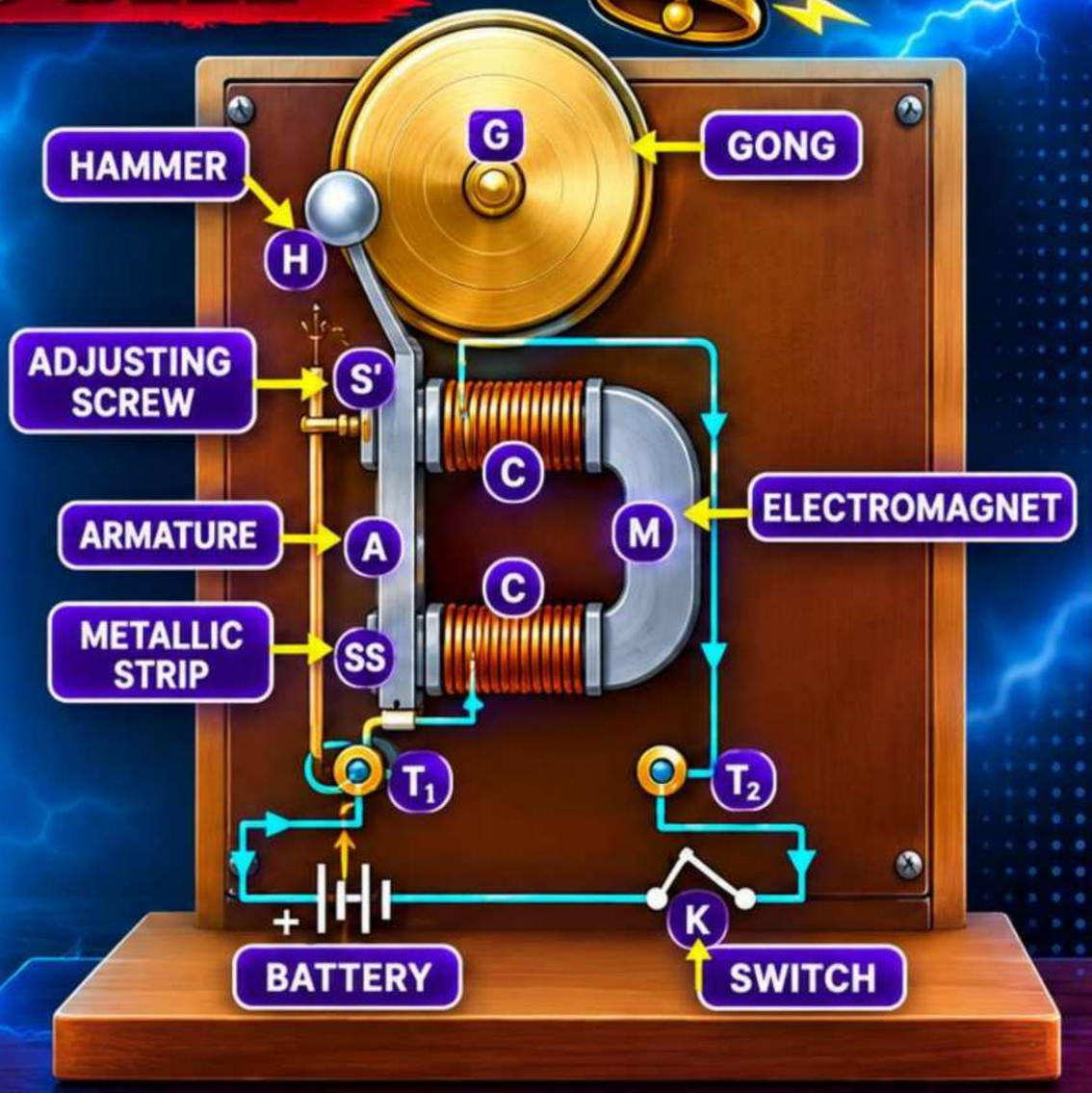
(iv) a gong G,

(v) a metallic spring strip SS,

(vi) an adjusting screw S',

(vii) a switch (or bell-push) K,

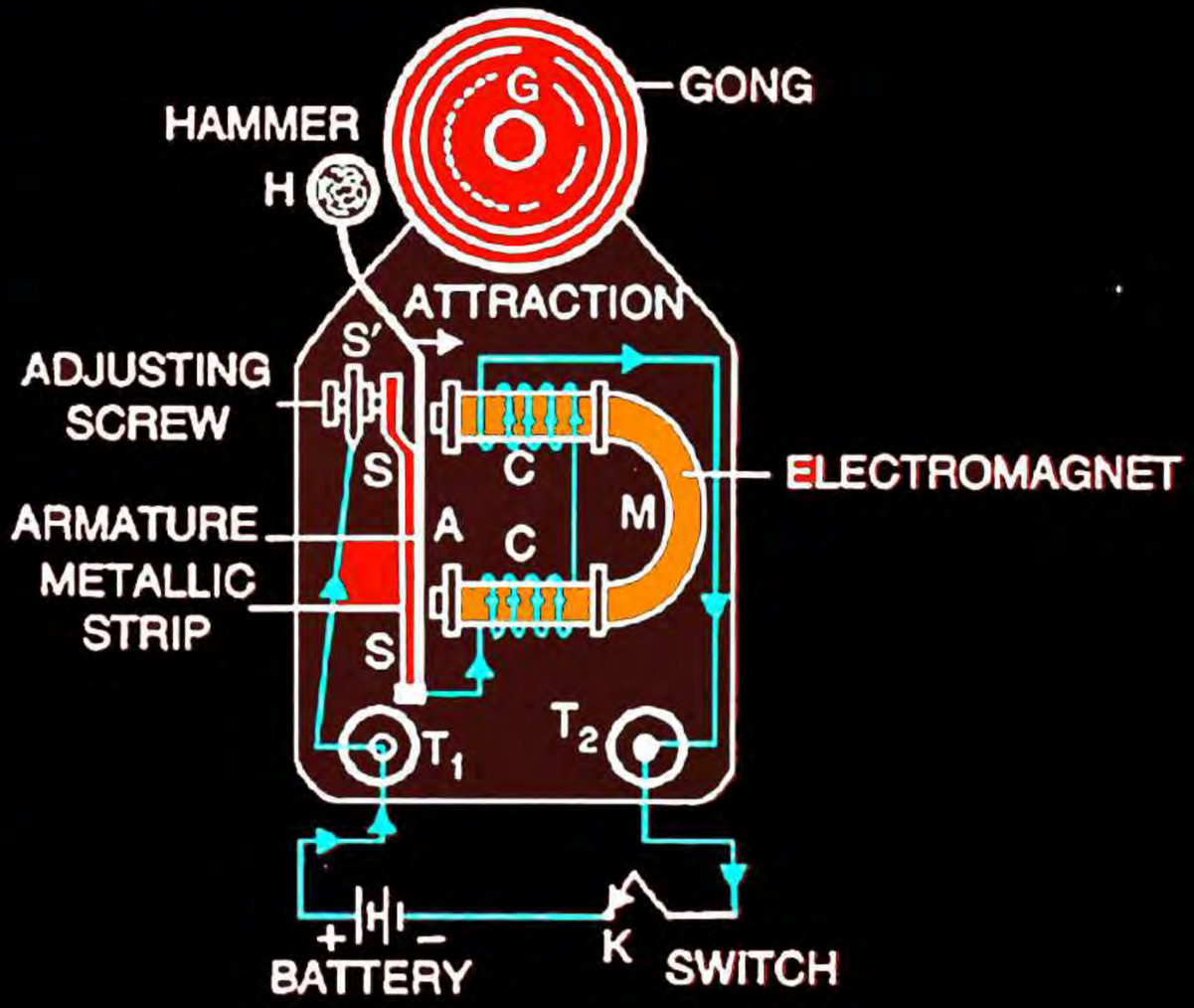
(viii) a battery.



USE OF ELECTROMAGNET IN AN ELECTRIC BELL

CONSTRUCTION AND WIRING

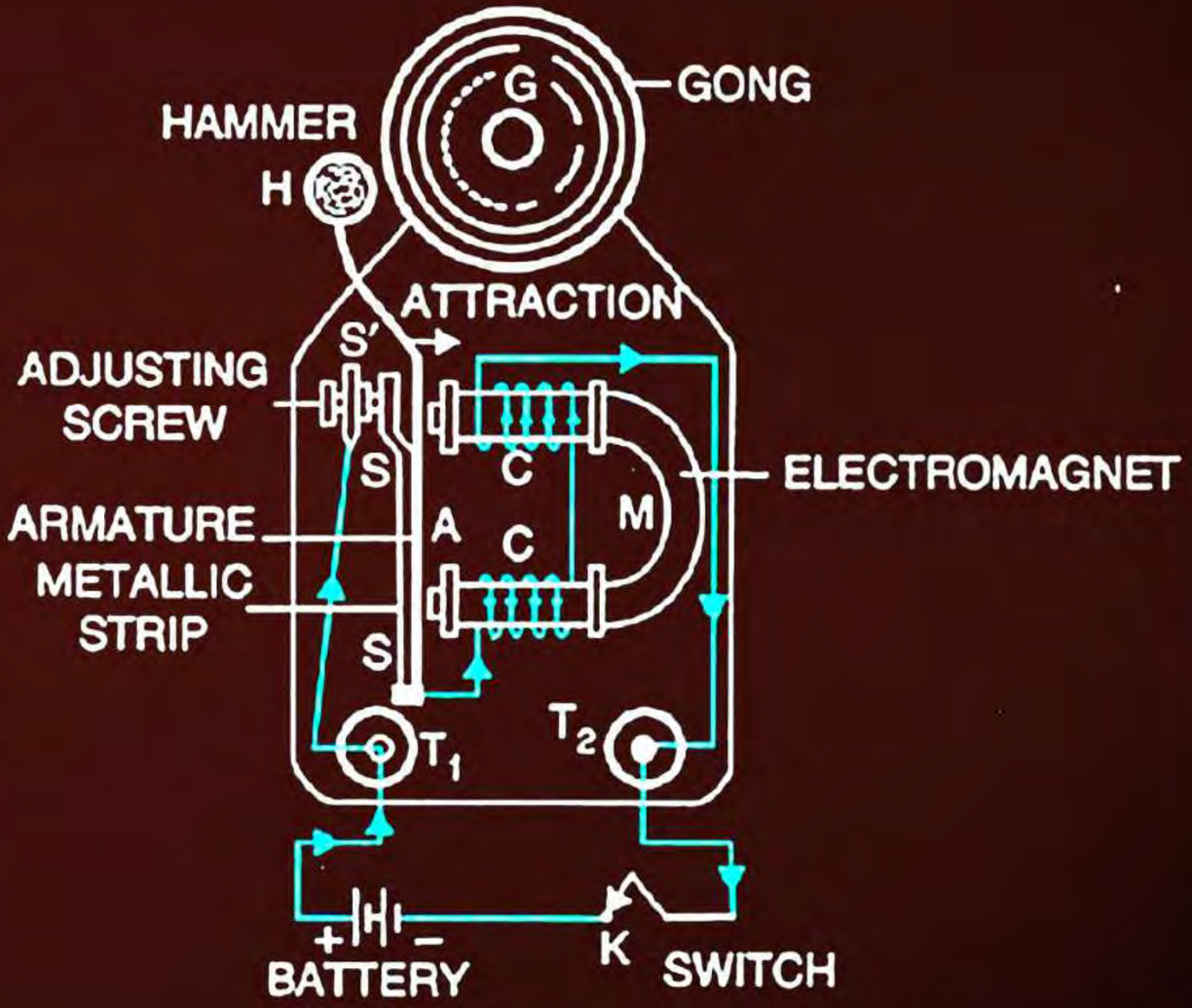
- Armature A is fixed to the spring strip SS, and the hammer H is attached to its upper end.
- When switch K is open, SS touches the adjusting screw S′, while a gap remains between A and the electromagnet poles.
- The coil CC is wound around the two arms of the electromagnet in opposite directions.
- One end of CC connects to T_1 through SS and S′; the other connects directly to T_2.
- A battery and switch K are connected in series across T_1 and T_2.



USE OF ELECTROMAGNET IN AN ELECTRIC BELL

WORKING AND FUNCTION OF EACH PART

- Switch K is pressed → Current flows through coil CC, magnetising the electromagnet.
- The electromagnet attracts armature A, causing hammer H to strike gong G.
- As A moves, SS loses contact with S′, breaking the circuit.
- Current stops → the electromagnet gets demagnetised.
- Due to the spring action of SS, armature A returns and touches S′ again.
- The circuit is completed again → electromagnet regains magnetism → A is attracted again.
- This make-and-break cycle repeats rapidly, making the hammer strike the gong continuously.



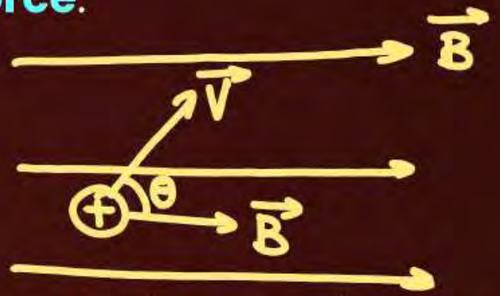
USE OF ELECTROMAGNET IN AN ELECTRIC BELL

- **Note:** If an a.c. source is used in place of battery, attraction of armature does not depend on the core of electromagnet will get magnetised, polarity of electromagnet, so the bell will still but the polarity at its ends will change. Since ring on pressing the switch K.

FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD

- Lorentz found that a charge moving in a magnetic field, in a direction other than the direction of magnetic field, experiences a force. It is called the **Lorentz force**.

$$F = qvB\sin\theta$$



- Since charge in motion constitutes a current, therefore a conductor carrying moving charges (or current) placed in a magnetic field, in direction other than the direction of magnetic field, will also experience a force and can produce motion in the conductor.

$$F = IlB\sin\theta$$

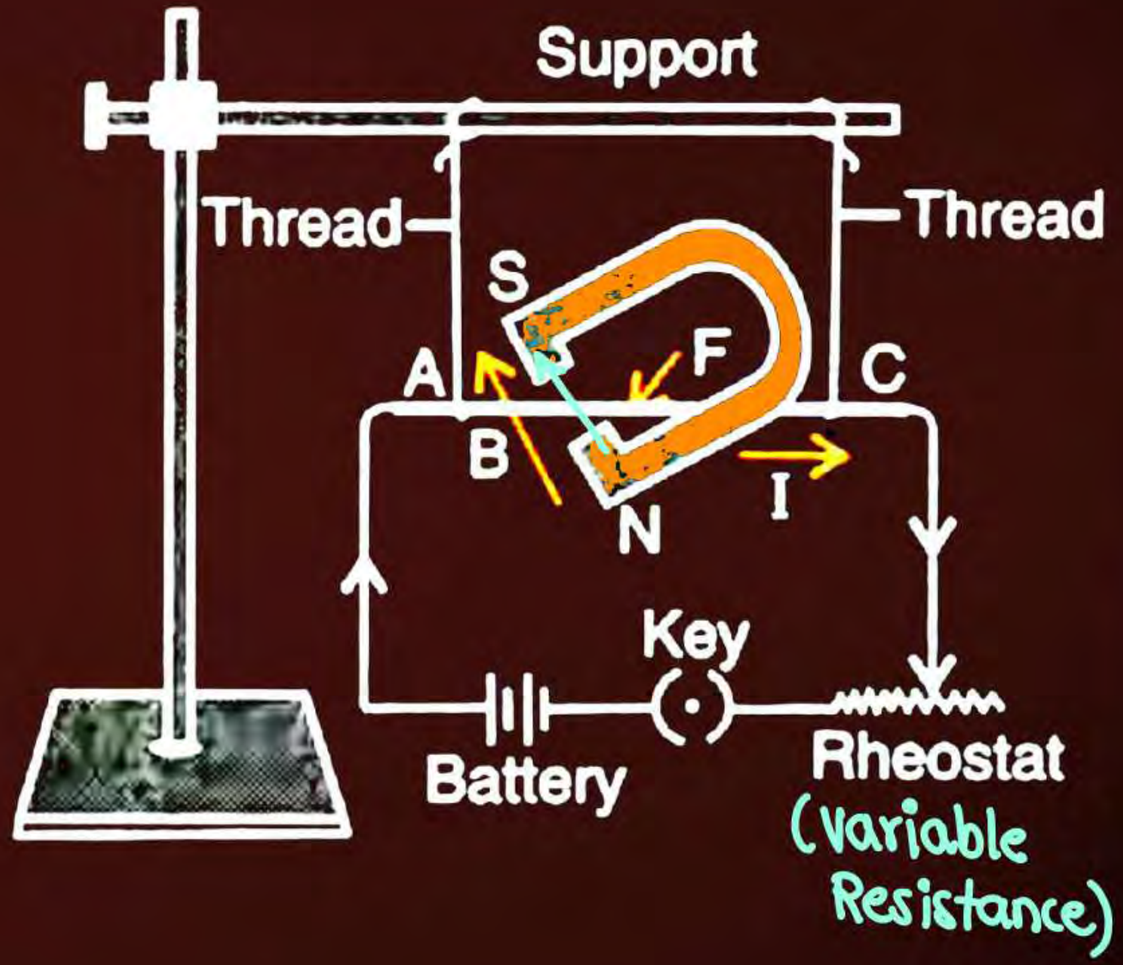


$\theta = 0^\circ, 180^\circ \Rightarrow F = 0$

$\theta = 90^\circ \Rightarrow F_{max}$

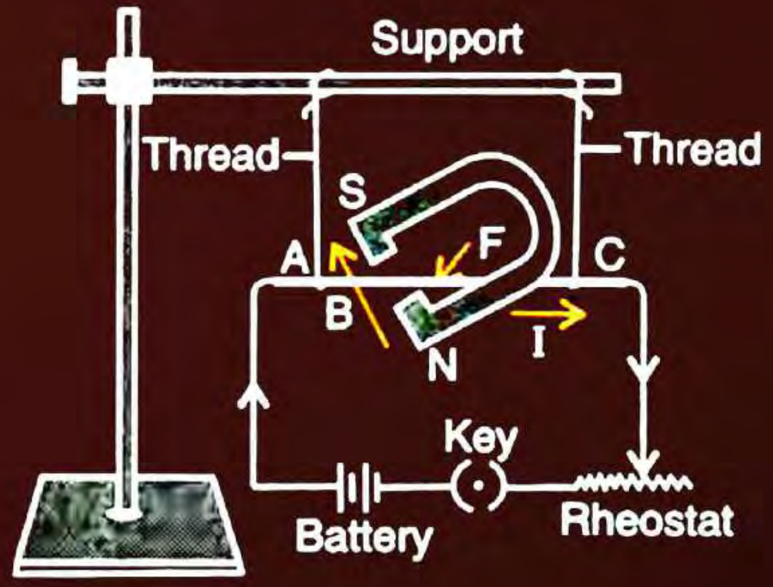
FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD

- **No current** → No force: If no current flows through the wire, no magnetic force acts on it, so it remains stationary.
- **Current in AC direction**: When current flows from A → C (X-direction) in magnetic field B, a force F acts perpendicular to both current and magnetic field. The wire moves outward (+Z direction).
- **Reverse current → Reverse force**: On reversing the battery terminals, the current direction reverses, so the force and motion also reverse → -Z direction (inward).
- **Reverse magnetic field → Reverse force**: On interchanging the N and S poles of the magnet, the magnetic field reverses, causing the force and motion to reverse → -Z direction.



FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD

Note: If wire is short enough and is placed between the poles of magnet with its length AC in Y direction such that the **current in it is in the direction parallel to the direction of magnetic field, no force** acts on the wire and it does not move.

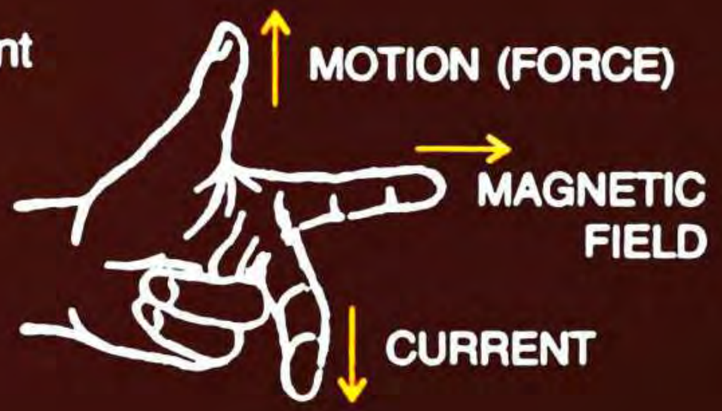


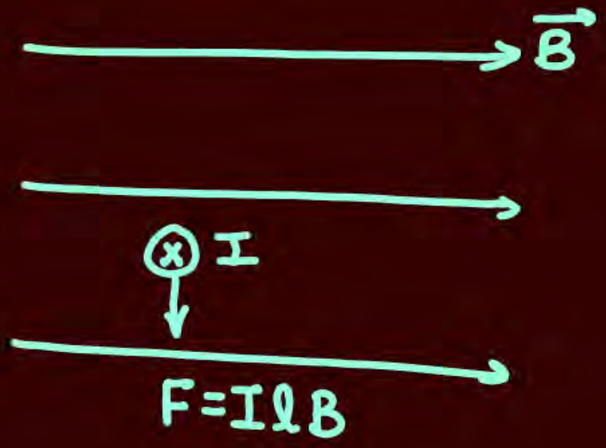
FLEMING'S LEFT HAND RULE FOR THE DIRECTION OF FORCE

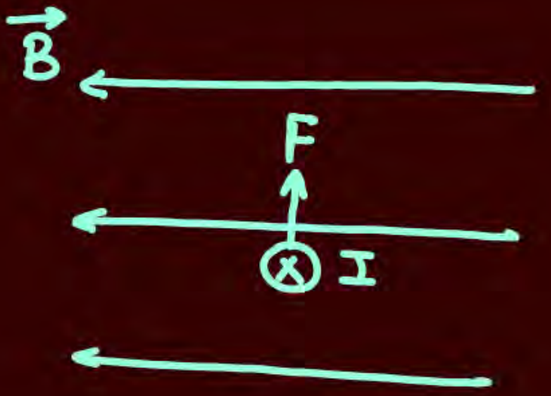
The direction of force on a current carrying conductor placed in a magnetic field is obtained by the Fleming's left hand rule.

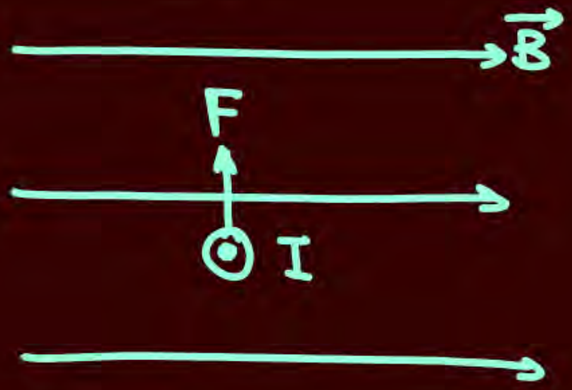
- **Stretch the forefinger, central finger and the thumb of your left hand mutually perpendicular to each other. If the forefinger indicates the direction of magnetic field and the central finger indicates the direction of current, then the thumb will indicate the direction of motion of conductor (i.e., force on conductor).**

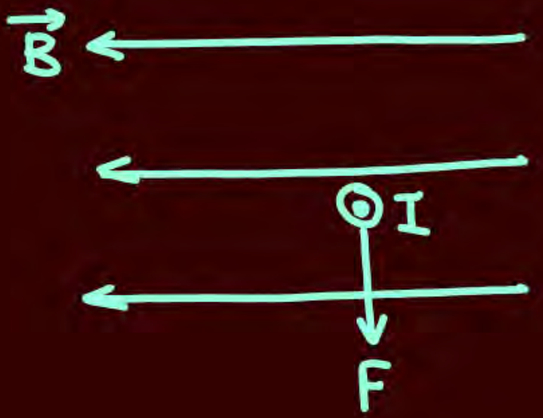
Central finger—Current
Fore finger—Field
Thumb—Motion

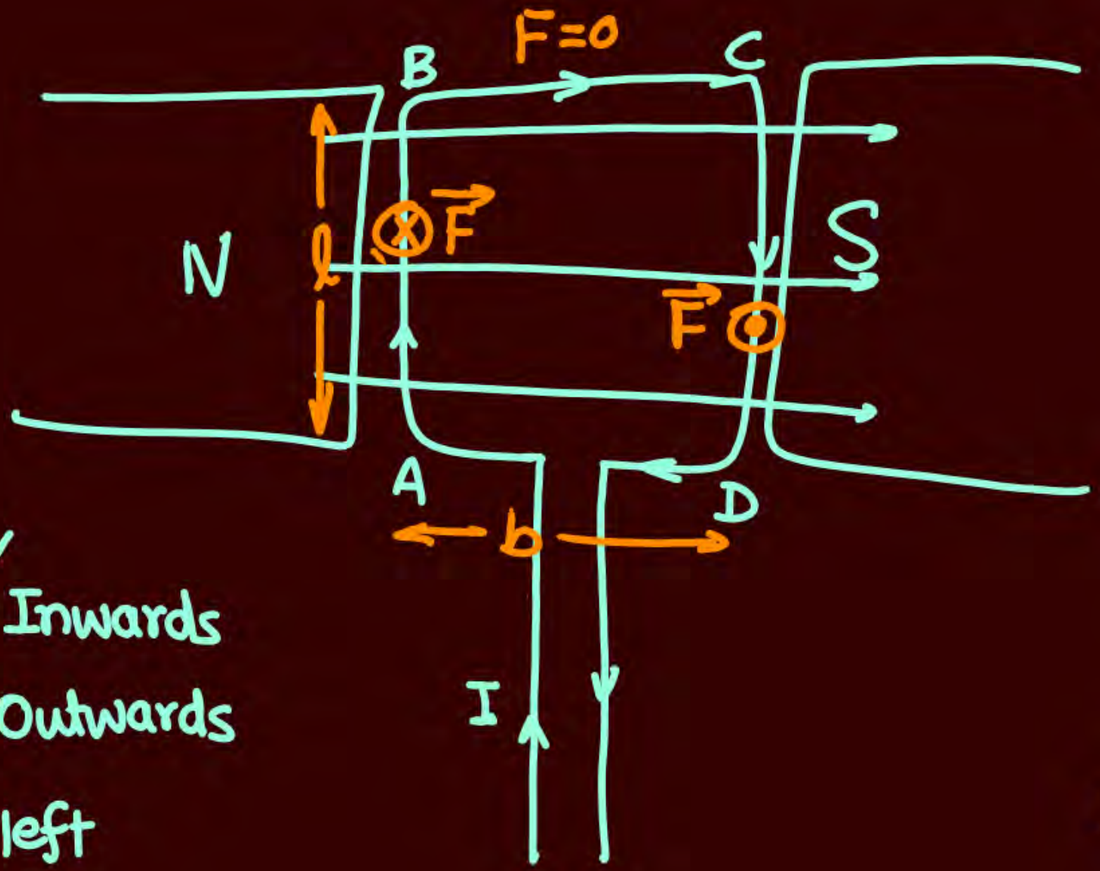



B
I
F=IlB


B
F
I


B
F
I


B
I
F



(a) Inwards

(b) Outwards

(c) left

(d) right

(e) Bail

Deflecting Torque or Couple

$$\tau = n \times F$$

$$= b \times (I l B \sin\theta)$$

$$= I (lb) B \sin\theta$$

$$\boxed{\tau = I A B \sin\theta}$$

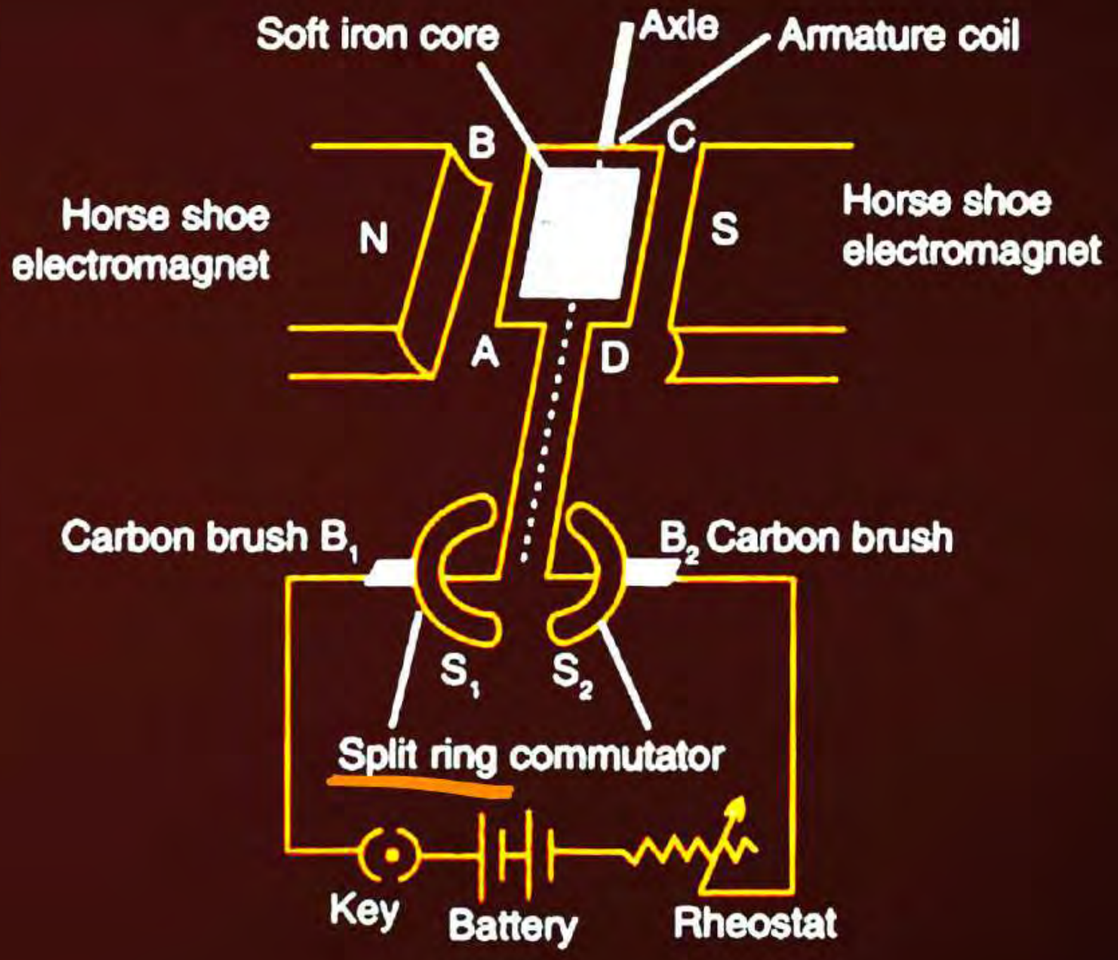
For N turns

$$\boxed{\tau = N I A B \sin\theta}$$

SIMPLE D.C. MOTOR

An electric motor is a device which converts the **electrical energy** into the **mechanical energy**.

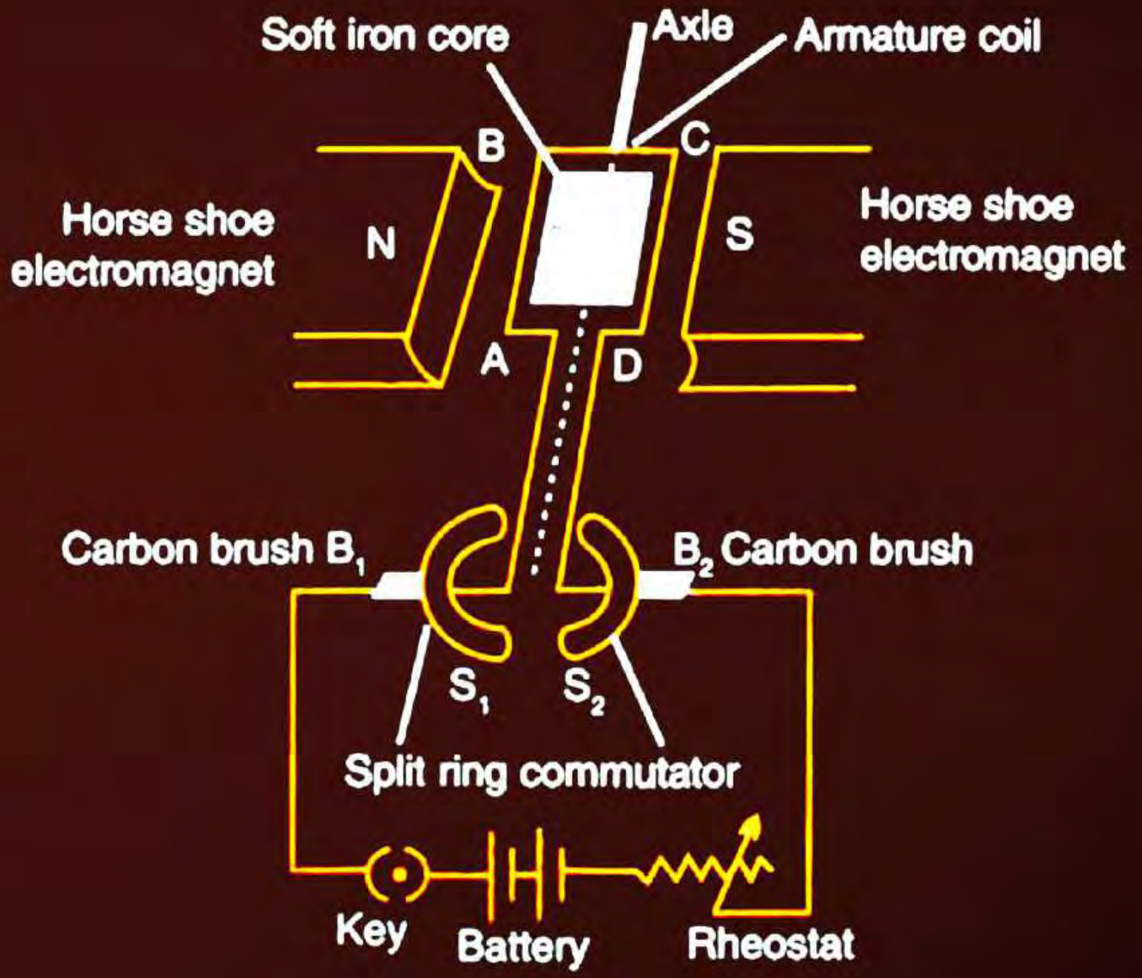
Principle: A d.c. motor works on the principle that when an electric current is passed through a conductor placed normally in a magnetic field, a force acts on the conductor as a result of which the conductor begins to move and thus mechanical energy (or work) is obtained. The direction of force on the conductor is obtained with the help of Fleming's left hand rule.



SIMPLE D.C. MOTOR

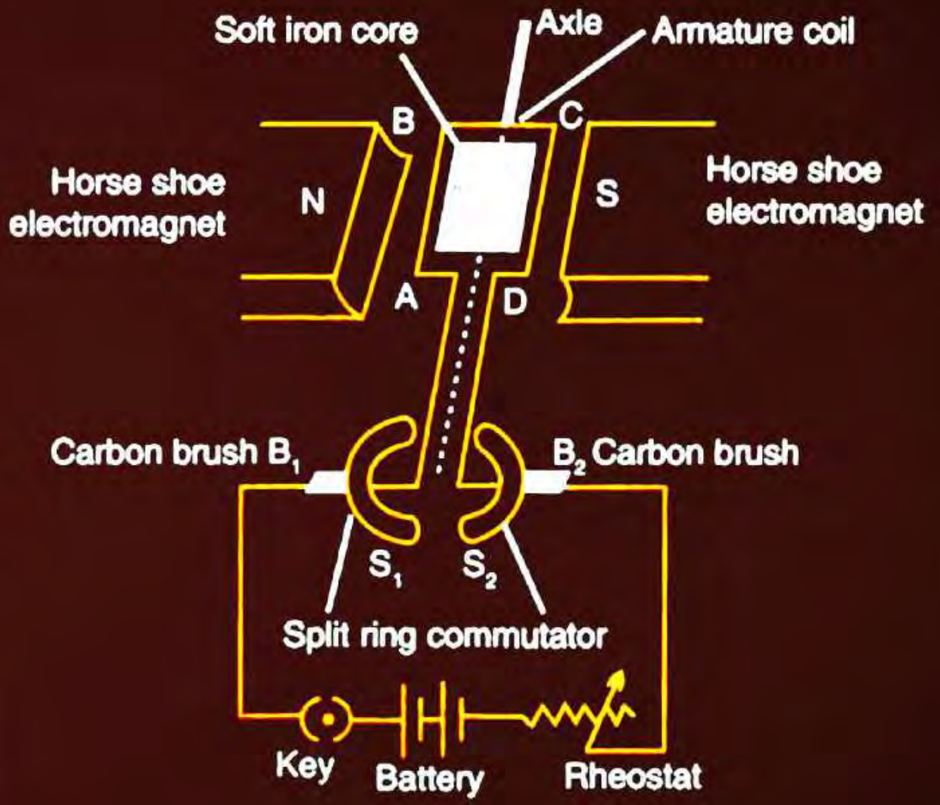
CONSTRUCTION AND ITS MAIN PARTS

- the **armature coil** ABCD mounted on an axle,
- the split parts S_1 and S_2 of a ring (i.e., a copper slip ring being divided in two parts S_1 an S_2) which is known as **split ring commutator**,
- a **pair of carbon (or copper) brushes** B_1 and B_2,
- a **horse-shoe electromagnet** NS, and
- a **d.c. source** (i.e., battery).



SIMPLE D.C. MOTOR

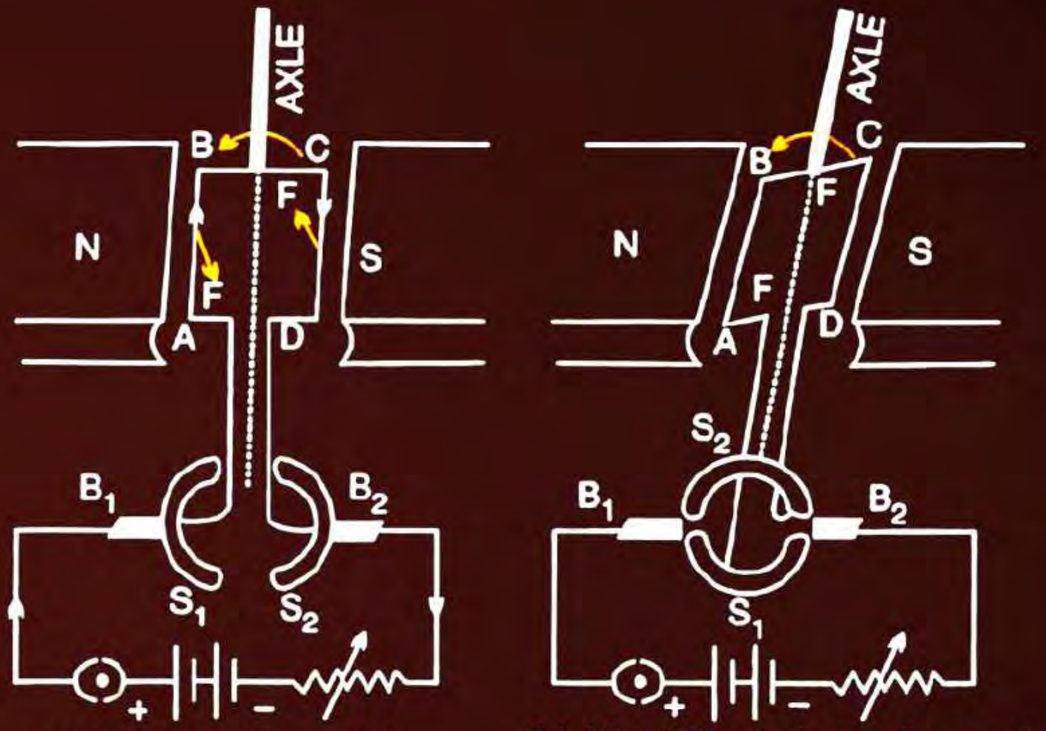
- The armature coil (ABCD) is wound on a soft iron core and placed between the north and south poles of a strong electromagnet.
- The coil can rotate freely around its axis.
- The ends A and D of the coil are connected to two halves of a split ring (S_1 and S_2).
- Brushes (B_1 and B_2) press lightly against these split rings - they are fixed, while the ring rotates with the coil.
- A battery (DC source) is connected across the brushes through a key and rheostat (for controlling current).
- When current flows and the coil rotates, the split rings keep changing contact with brushes - ensuring current direction reverses every half turn.
- A wheel or machine part can be attached to the coil's axle to perform useful work (as in a motor).



SIMPLE D.C. MOTOR

Working:

- When the key is closed, current flows in the coil in the direction ABCD.
- The arms BC and DA are parallel to the magnetic field → no force acts on them.
- The arms AB and CD are perpendicular to the magnetic field → each experiences a force (F = IBl).
- By Fleming's Left-Hand Rule:
  - Force on AB is inward (into the paper).
  - Force on CD is outward (out of the paper).
- These equal and opposite forces form an anticlockwise couple, → making the coil rotate so that AB moves in and CD comes out.



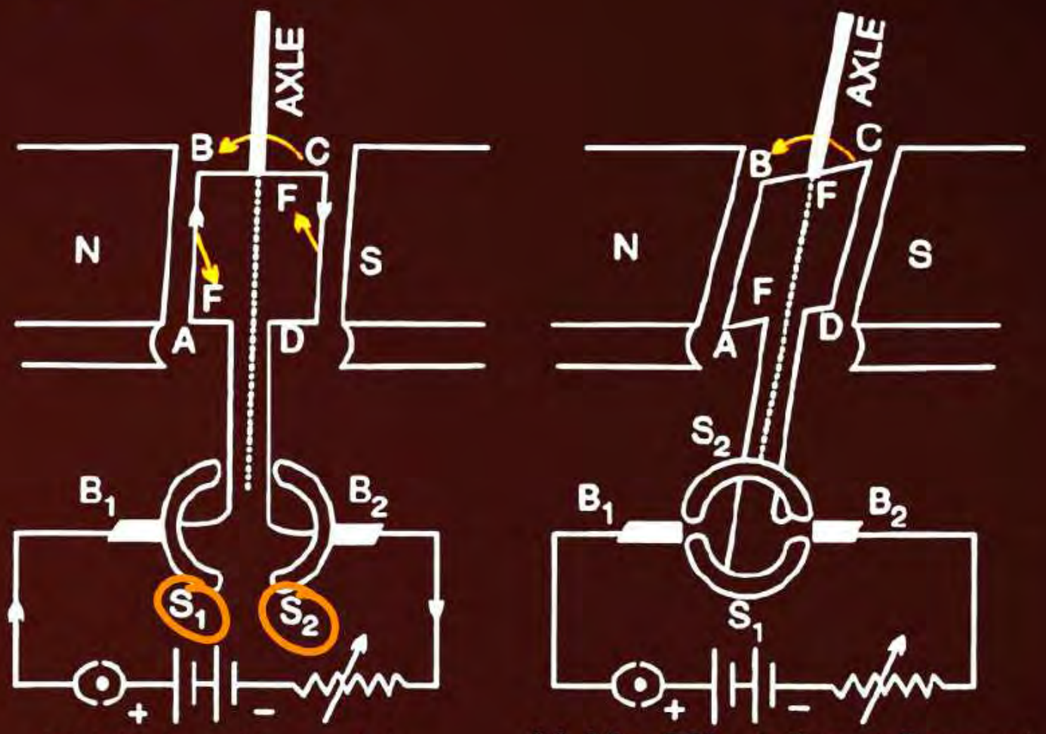
(a) Initial position, coil in plane of magnetic field

(b) After 90° rotation, coil normal to the plane of magnetic field.

SIMPLE D.C. MOTOR

- After rotating $90°$, the coil becomes vertical - arm CD is out, and AB is in.
- At this position, the brushes lose contact with the split rings (S_1 and S_2).
- So, no current flows through the coil → no magnetic force acts on it.
- But due to its inertia, the coil keeps rotating and does not stop in this position.

A→B→C→D



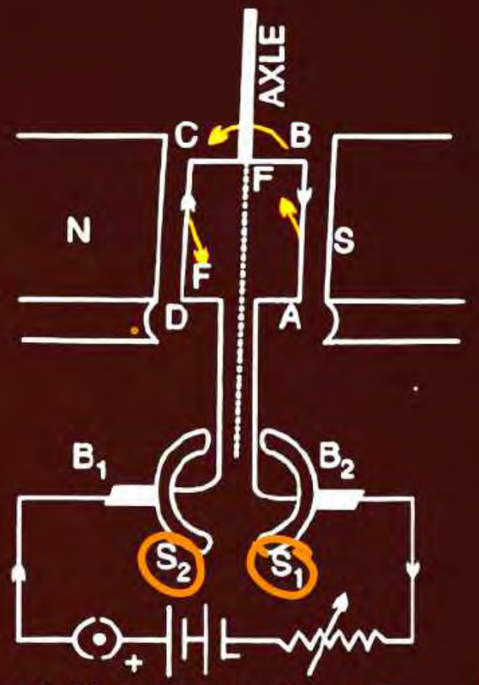
(a) Initial position, coil in plane of magnetic field

(b) After 90° rotation, coil normal to the plane of magnetic field.

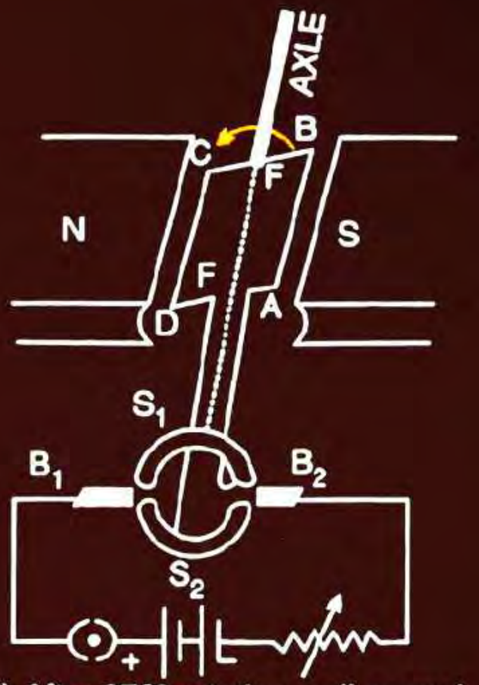
SIMPLE D.C. MOTOR

D→C→B→A

- As the coil moves past the vertical position, the split rings change contact:
  - S_1 touches brush B_2,
  - S_2 touches brush B_1.
- The current direction reverses → now flows as DCBA.
- Arms CB and AD are parallel to the magnetic field → no force on them.
- On arm DC, force acts inward (into the paper).
- On arm BA, force acts outward (out of the paper).
- These equal and opposite forces again form an anticlockwise couple, → keeping the coil rotating in the same direction.
- The force is maximum when the coil has turned $180°$ from its starting position.



(c) After 180° rotation, coil in plane of magnetic field



(d) After 270° rotation, coil normal to the plane of magnetic field.



SIMPLE D.C. MOTOR

- After turning $270°$, the coil again becomes vertical.
- At this position, the brushes lose contact with the split rings → no current flows through the coil.
- Hence, no magnetic force acts on the coil at that instant.
- But due to inertia of motion, the coil keeps rotating and soon returns to its initial position.



Working of a DC Motor (Step-by-Step Summary)

- Initial Position (Key Closed, Current ABCD):
- Current flows in the coil ABCD.
- Arms AB and CD are perpendicular to the magnetic field → each experiences a force (F= IBl).
- By Fleming's Left-Hand Rule:
  - Force on AB → inward (into the paper).
  - Force on CD → outward (out of the paper).
- These opposite forces form an anticlockwise couple, rotating the coil.
- After 90° Rotation (Vertical Position):
- Coil becomes vertical → brushes lose contact with the split rings.
- No current and hence no force acts on the coil.
- Due to inertia, the coil keeps rotating.
- Beyond 90° (After 180° Rotation):
- Split rings change contact with brushes.
- Current now flows DCBA (direction reversed).
- Arms DC and BA again experience equal and opposite forces, → forming another anticlockwise couple.
- Coil continues rotating in the same direction.
- After 270° Rotation (Second Vertical Position):
- Coil again becomes vertical, brushes lose contact momentarily.
- No current flows, but due to inertia, the coil continues rotating.
- After 360° Rotation (Initial Position Reached):
- Coil regains its original position.
- The same process repeats continuously, → making the coil (and attached wheel) rotate steadily.



NOTE

- Deflecting Couple (Turning Effect) $\tau = NIABSin\theta$
- The deflecting couple is the twisting force that makes the coil rotate.
- It is maximum when the plane of the coil is parallel to the magnetic field. $\theta = 90^\circ \Rightarrow \tau_{max} = NIAB$
- It is zero (or minimum) when the plane of the coil is perpendicular to the magnetic field.

$\theta = 0^\circ$ or $180^\circ \Rightarrow \tau = 0$

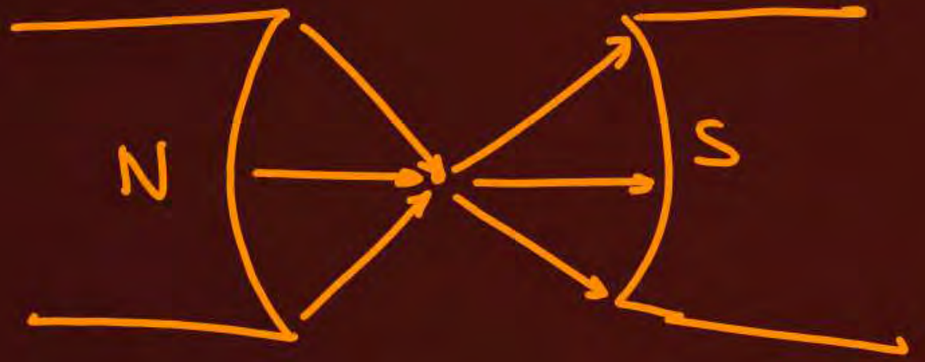
- Speed of Rotation
- The speed of the motor depends on the magnitude of the deflecting couple.
- Deflecting couple = n I A B, where:

n = number of turns in coil
I = current in coil
A = area of coil
B = magnetic field strength

- Therefore, the **speed of rotation $\propto n$ I A B**
- → i.e., it increases with stronger field, larger coil, more turns, or higher current.

NOTE

To increase the strength of magnetic field, a **soft iron core** is inserted within the coil. The **insertion of core makes the magnetic field strong and radial**. Due to radial field, the couple acting on the coil remains almost constant during the rotation of coil, except in the vertical position of coil when the couple is zero.



$\theta = 90^\circ$

$\tau = NIAB$

SIMPLE D.C. MOTOR

- **FUNCTION OF SPLIT RING COMMUTATOR:**

  Whenever the coil passes from the vertical position, the direction of current through the coil has to be reversed, so that the coil continues to rotate in the same direction. This is achieved by the two split parts S_1 and S_2 of a conducting ring which reverses the direction of current in the coil. They are called the split ring commutator.

- **TRANSFORMATION OF ENERGY IN A D.C. MOTOR:**

  In a d.c. motor, the electrical energy supplied by the battery to the coil is transferred to the mechanical energy which rotates the coil.

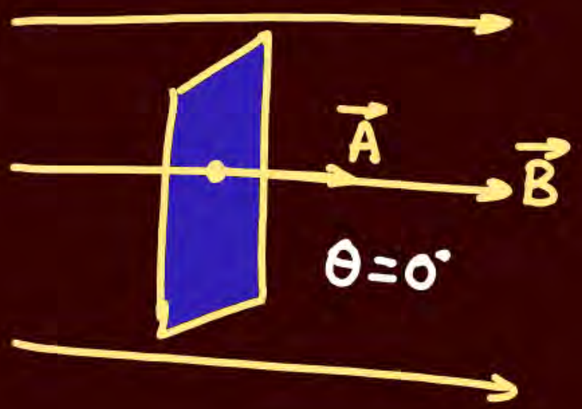


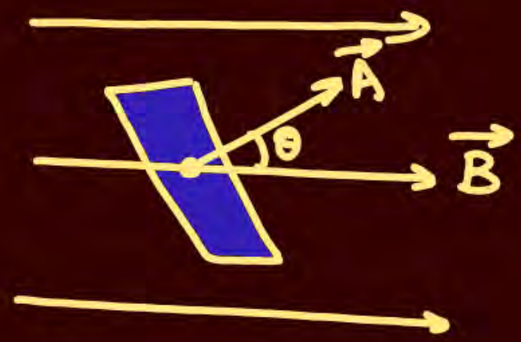
QUESTIONS

In an electric motor, the energy transformation is:

1. from electrical to chemical
2. from chemical to light
3. from mechanical to electrical
4. from electrical to mechanical

Magnetic Flux





$$\phi_B = BA\cos\theta$$

Magnetic Flux → No of Magnetic field lines passing through coil perpendicularly

$$\phi_B = BA\cos 0^\circ = BA$$

(SI) Tm^2 or Weber

(CGS) G cm^2 or Maxwel

$$1\,Wb = 1\,Tm^2 = 10^4\,G \cdot 10^4\,cm^2 = 10^8\,M_x$$

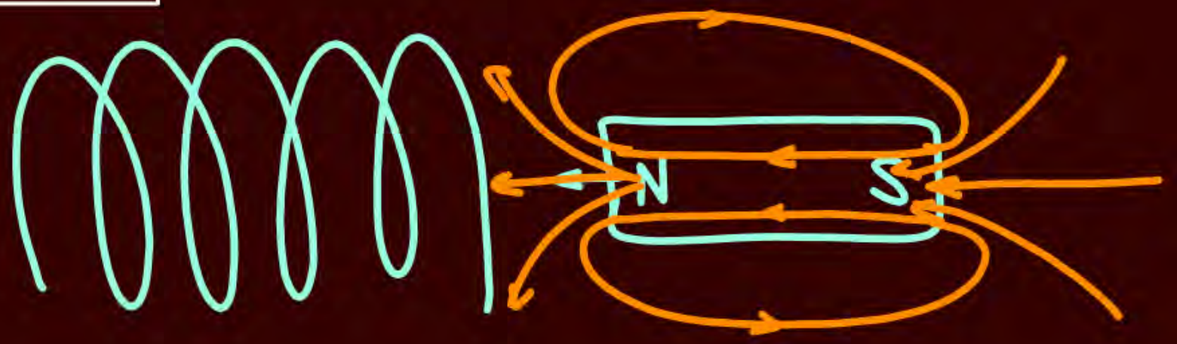
Electromagnetic Induction

Change in Magnetic flux of coil $\longrightarrow$ $E_{induced}$

$\downarrow$

$I_{induced}$

Faraday's Law



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.



Thank You